



DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Experiment No.: 2

Interfacing LPC 2148 to seven segment display and display a count from 0 to 99.

Name: _____

Roll No.: _____ *Batch:* _____

Date of Performance: _____

Date of Assessment: _____

Particulars	Marks
Marks for Regularity (05)	
Marks for Performance (15)	
Understanding (Viva) (05)	
Total (25)	
Signature of Course Teacher	

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Experiment: 2

AIM: Interfacing LPC 2148 to seven segment display and display a count from 0 to 9 with suitable delay.

OBJECTIVE:

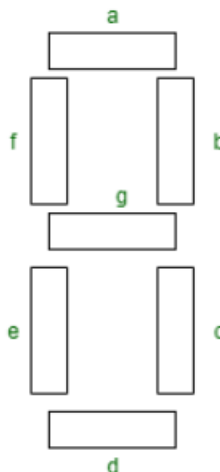
1. To understand the working of a seven-segment display (common anode / common cathode).
2. To interface a seven-segment display with LPC2148 GPIO pins.
3. To learn how to send binary patterns to display digits (0-9).

APPARATUS:

1. LPC2148 module development board.
2. Connecting wire.
3. Adapter.

THEORY:

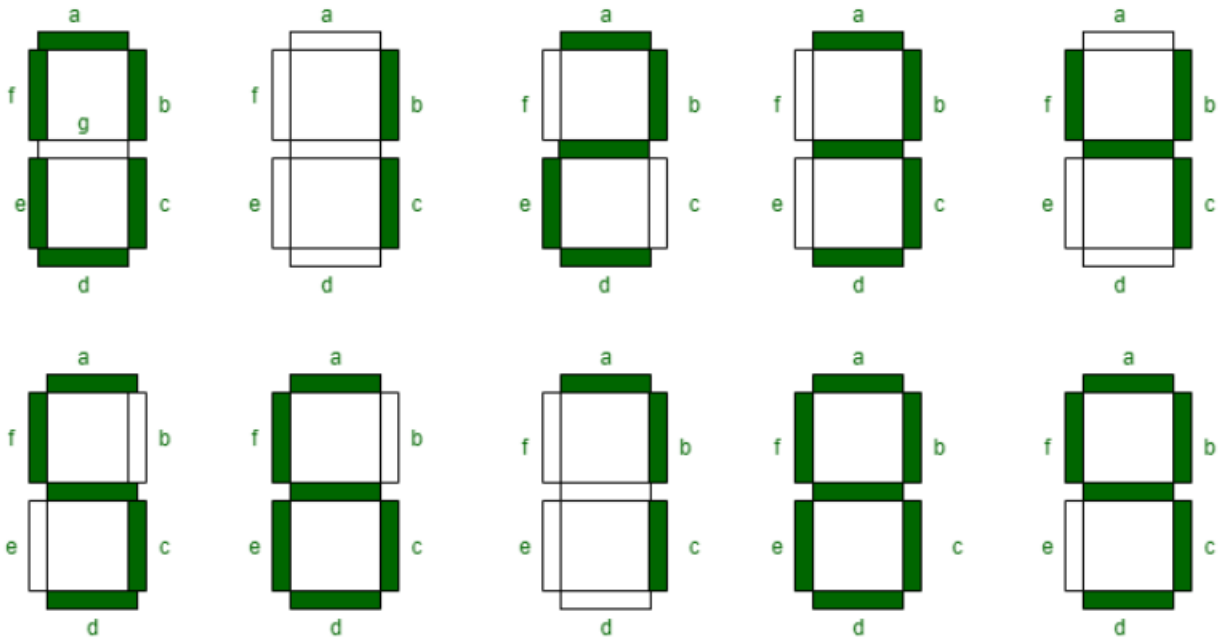
Seven segment displays are the output display device that provides a way to display information in the form of images or text or decimal numbers which is an alternative to the more complex dot matrix displays. It is widely used in digital clocks, basic calculators, electronic meters, and other electronic devices that display numerical information. It consists of seven segments of light-emitting diodes ([LEDs](#)) which are assembled like numerical 8.



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Working of Seven Segment Displays:

The number 8 is displayed when the power is given to all the segments and if you disconnect the power for 'g', then it displays the number 0. In a seven-segment display, power (or voltage) at different pins can be applied at the same time, so we can form combinations of display numerical from 0 to 9. Since seven-segment displays cannot form alphabets like X and Z, so it cannot be used for the alphabet and they can be used only for displaying decimal numerical magnitudes. However, seven-segment displays can form alphabets A, B, C, D, E, and F, so they can also be used for representing each display unit is usually has a dot point (DP).



Truth Table: We can produce a truth table for each decimal digit.

Boolean expressions for each decimal digit that requires respective light-emitting diodes (LEDs) are ON or OFF. The number of segments used by digit: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 are 6, 2, 5, 5, 4, 5, 6, 3, 7, and 6 respectively. Seven segment displays must be controlled by other external devices where different types of microcontrollers are useful to communicate with these external devices, like switches, keypads, and memory.

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Decimal Digit	Individual Segments Illuminated						
	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	1	0	0	0	0
2	1	1	0	1	1	0	1
3	1	1	1	1	0	0	1
4	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1
6	1	0	1	1	1	1	1
7	1	1	1	0	0	0	0
8	1	1	1	1	1	1	1
9	1	1	1	1	0	1	1

Types of Seven Segment Displays:

According to the type of application, there are two types of configurations of seven-segment displays: common anode display and common cathode display.

- In common cathode seven segment displays, all the cathode connections of LED segments are connected together to logic 0 or ground. We use logic 1 through a current limiting resistor to forward bias the individual anode terminals a to g.
- Whereas all the anode connections of the LED segments are connected together to logic 1 in a common anode seven segment display. We use logic 0 through a current limiting resistor to the cathode of a particular segment a to g.

Common anode seven segment displays are more popular than cathode seven segment displays because logic circuits can sink more current than they can source and it is the same as connecting LEDs in reverse.

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Advantages of Seven Segment Displays:

- **Simplicity:** Seven Section Presentations are straightforward and simple to use since they just showcase mathematical digits (0-9) and a couple of characters like A-F for hexadecimal numbers.
- **Cost-viable:** Seven Section Presentations are generally modest and require less parts to work than different sorts of showcases like LCDs or OLEDs.
- **High perceivability:** Seven Portion Presentations have high perceivability even in low light circumstances as they are intended to emanate splendid, high-contrast light in a particular example that is not difficult to peruse.
- **Durability:** Seven Section Presentations are strong and sturdy since they are produced using materials that are impervious to temperature changes and mechanical pressure.

Common Cathode (CC) Configuration:

In the common cathode configuration, all the cathodes (negative terminals) of the seven LEDs are connected to a common ground. Display segments are controlled independently by applying a HIGH (Logic 1) voltage to the anode of the particular LED. On applying the HIGH signal to an anode, current flows from the anode to the common cathode, and accordingly, the respective segment lights up.

Common Anode (CA) Configuration:

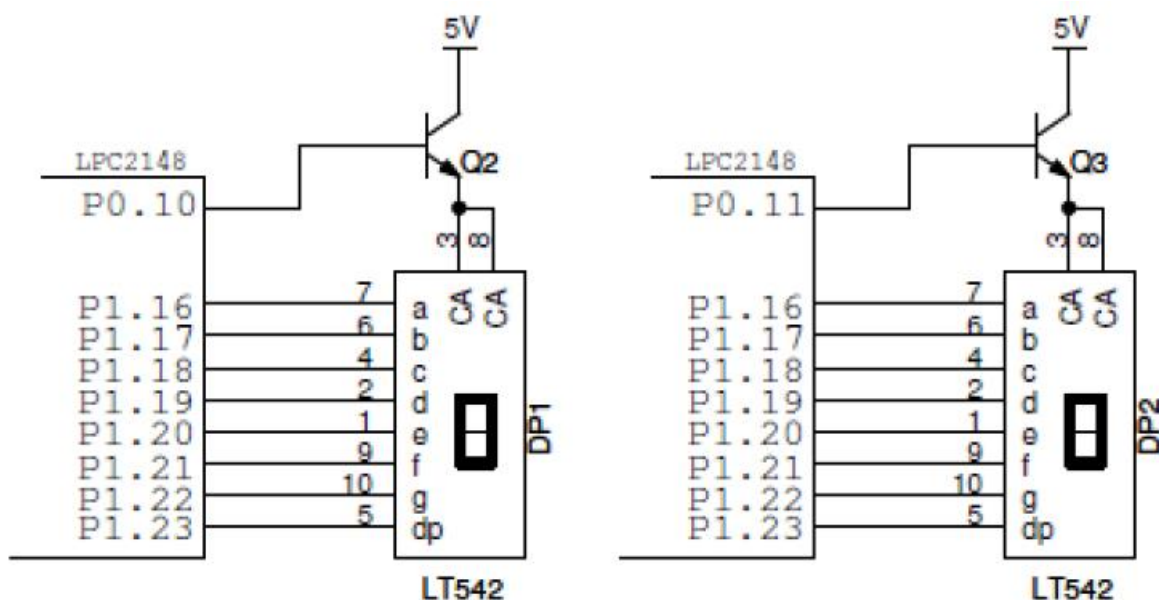
In an anode common configuration, the positive terminals of the seven LEDs are connected together to a common positive voltage. Typically, this is VCC. Individual segments are turned on by grounding the respective cathode. A LOW, or logic 0, voltage at a cathode sends current flowing from the CA, through the LED segment, to ground and turns it on.

REAL-WORLD APPLICATIONS:

- Digital clocks
- Clock radios
- Calculators
- Wristwatches
- Speedometers

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INTERFACING DIAGRAM:



LCD INTERFACING DETAILS:

Seven Segment	Pin Details	
	LPC2148	Description
A	P1.16	Seven Segment A
B	P1.17	Seven Segment B
C	P1.18	Seven Segment C
D	P1.19	Seven Segment D
E	P1.20	Seven Segment E
F	P1.21	Seven Segment F
G	P1.22	Seven Segment G
DP	P1.23	Seven Segment DP
SEG_SEL1	P0.10	Seven segment select 1
SEG_SEL2	P0.11	Seven segment select 2



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CONCLUSION:
